

# Compiler Register Allocation

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## DA Programming Project II

Group T9G2 / Spring 2026

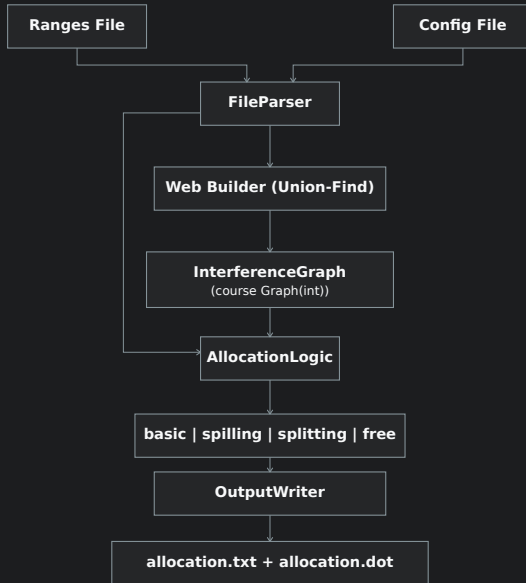


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`make test`

`make docs`

`./register_alloc -b ...`



# Spilling and Splitting

## Bounded spilling

Colored Register Allocation  
2 register(s), 1 memory web(s)

spills: web0

Legend

orange border = split-derived

M = memory

same fill = same register

web0  
a  
M  
1+,2,3-

web1  
b  
r1  
1+,2,3-

web2  
c  
r0  
1+,2,3-

## Bounded splitting

Colored Register Allocation  
2 register(s), 0 memory web(s)

splits: web0 -> web0,web3

Legend

orange border = split-derived

M = memory

same fill = same register

web0  
a  
r0  
1+,2

web1  
b  
r1  
1+,2,3,4-

web2  
c  
r0  
3+,4,5,6-

web3  
a  
r2  
5,6

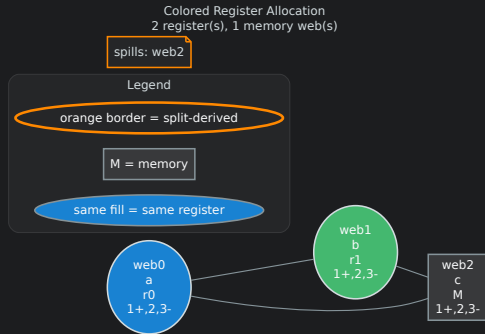
Splitting preserves original markers. 1+,2,5,6- splits into 1+,2 and 5,6- — no fabricated 2- or 5+.

Metadata: spills: N / splits: N.

# Custom Free Algorithm

## Hybrid strategy

- edgeless graphs: 1 register
- complete graphs: K colored, rest spilled
- bipartite graphs: BFS 2-coloring
- general: DSatur greedy coloring
- small/medium: branch-and-bound refines spills



Invariant: interfering colored webs never share a register. Memory only for uncolored webs.

# Demo and Testing

## Inputs for the demo

- `complex_allocation.txt` + `free4.txt`: dense allocator stress test
- `splitting_showcase.txt` + `splitting2.txt`: split reduces K from 3 to 2
- `spilling5.txt`: bounded spilling in a dense graph

```
./register_alloc -b
  inputs/advanced/ranges/
complex_allocation.txt
  inputs/advanced/registers/free4.txt
  alloc.txt
```

`make test` — 76 unit + 9 integration (passes clean). DOT graphs written alongside allocation output.

# Complexities

## Input & graph building

**Parse:**  $O(LP)$

**Build  
webs:**  $O(VR^2P \log P)$

**Interference:**  $O(W^2P + EW)$

## Coloring algorithms

**Basic:**  $O(W(W^2 + E))$

**Spilling:**  $O((S + 1)W(W^2 + E))$

**Splitting:**  $O(SQ(W^2P + EW + W(W^2 + E)))$

**Free:**  $O(W^2 + E \log W)$  (heuristic)

**Variables:**  $W$ =webs,  $E$ =edges,  $P$ =points,  $R$ =ranges/var,  $S$ =recovery bound,  $Q$ =split candidates,  $L$ =live ranges.